

# OpenAI's Whisper: A New Era in Audio Transcriptions

Whisper, OpenAI's latest speech recognition model, opens doors to seamless speech-to-text transcription, revolutionising communication across languages.



Today, generative AI has become the hottest topic and has the highest buzz in the field of technology. Everyone wants to know what it is and how to use it the best way they can. GenAI is basically the subfield of artificial intelligence designed to create new content, whether it be images, text, audio, or any other form of binary or non-binary data. This is in accordance with the data upon which the model has been trained. Today, OpenAI's ChatGPT is one of the most popular generative AI models that everyone is interested in. OpenAI has also released another amazing model, known as Whisper, which we will focus on here.

Whisper is an open source machine learning model that can recognise and transcribe speech, and can work with a wide range of languages. Given an audio input, it can detect the language, generate the transcription in that language, and then translate it into English.

Speech recognition has been a topic of research for a long time now. The earliest methods used techniques such as time warping and Hidden Markov Models (HMM). Eventually, like most applications of data science, deep learning approaches became prevalent in this field. With the help of Big Data technologies, deep learning proved to be very effective. Convolutional neural networks (CNNs) and recurrent neural networks (RNNs) were used majorly. The Transformers library introduced by Google disrupted most machine learning tasks and applications as it proved to be incredibly useful. Whisper has been built using this library, which is based on the encoder-decoder format. It can be said that Whisper is a major milestone in the field of speech-to-text transcriptions.

So let's learn how to use Whisper.

First, let us install Whisper. This can be done using the following command:

```
pip install whisper
```

```
(base) Jishnus-MacBook-Air:~ jishnusaurav$ pip install
whisper
Collecting whisper
Downloading whisper-1.1.10.tar.gz (42 kB)
Preparing metadata (setup.py) ... done
Requirement already satisfied: six in ./anaconda3/lib/
python3.11/site-packages (from whisper) (1.16.0)
Building wheels for collected packages: whisper
Building wheel for whisper (setup.py) ... done
Created wheel for whisper: filename=whisper-1.1.10-py3-none-
any.whl size=41121 sha256=8c8051947499458bc06c3b1ab1cff2f3005
604624c1fca58271356f596241070
Stored in directory: /Users/jishnusaurav/Library/Caches/pip/
wheels/21/65/ee/4e6672aabfa486d3341a39a04f8f7c77e5156149299
b5a7d0
```

```
Successfully built whisper
Installing collected packages: whisper
Successfully installed whisper-1.1.10
```

Sometimes, you may encounter an error. In that case, you can use the command shown in Figure 1.

Next, we need to install *ffmpeg*. If you are using a Mac,